A

Project Report

On

PhishShield AI – Intelligent Phishing Detection & Prevention System

Submitted by

Nishad Rahul Ramesh

Roll No.: 25242

MCA–I

SEM–II

Under the guidance of

Dr. Ramesh D. Jadhav

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Sinhgad Institutes

Sinhgad Technical Education Society's

Sinhgad Institute of Management

Vadgaon Bk Pune 411041

(Affiliated to SPPU Pune & Approved by AICTE New Delhi)

Date:

CERTIFICATE

This is to certify that Mr. Nishad Rahul Ramesh has successfully completed his project work entitled **"PhishShield AI – Intelligent Phishing Detection & Prevention System"** in partial fulfillment of MCA – I SEM –I Mini Project for the year 2025-2026. He/she has worked under our guidance and direction.

Dr. Ramesh D. Jadhav	Dr. Chandrani Singh
Project Guide	**Director, SIOM-MCA**

Date:

Place: Pune

DECLARATION

I certify that the work contained in this report is original and has been done by me under the guidance of my guide.

- The work has not been submitted to any other Institute for any degree or diploma.
- I have followed the guidelines provided by the Institute in preparing the report.
- I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
- Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the report and giving their details in the references.

Name and Signature of Project Team Members:

Sr. No.	Roll No.	Name of students	Signature of students
1	25242	**Nishad Rahul Ramesh**	

ACKNOWLEDGEMENT

It is very difficult task to acknowledge all those who have been of tremendous help in this project. I would like to thank my respected guide **Dr. Ramesh D Jadhav** for providing me necessary facilities to complete my project and for their guidance and encouragement in completing my project successfully without which it wouldn't be possible. I wish to convey my special thanks and immeasurable feelings of gratitude towards **Dr. Chandrani Singh, Director, SIOM-MCA.** I wish to convey my special thanks to all teaching and non-teaching staff members of **Sinhgad Institute of Management, Pune** for their support.

Thank You

Yours Sincerely,

Nishad Rahul Ramesh

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CHAPTER 1: INTRODUCTION

1.1 Abstract

The rapid growth of internet usage and digital communication has led to an increasing number of cyber threats, among which phishing attacks are one of the most common and dangerous. Phishing attacks attempt to deceive users into revealing sensitive information such as login credentials, financial details, or personal data through malicious websites, emails, or messages that mimic legitimate sources. Due to the increasing sophistication of phishing techniques, traditional manual detection methods are often ineffective and require intelligent automated solutions.

This project presents the design and development of PhishShield AI – an Intelligent Phishing Detection and Prevention System, which provides a web-based platform for detecting phishing attacks through automated analysis of suspicious URLs and email content. The system integrates machine learning techniques, blacklist verification, and rule-based analysis to identify potential phishing threats and calculate a risk score that helps users determine whether the content is safe or malicious.

The system allows users to scan suspicious URLs or email content, view detection reports, interact with an AI chatbot for cybersecurity guidance, and provide feedback regarding system performance. Administrators can monitor scan activities, manage blacklist and whitelist domains, analyze threat statistics, and generate reports through a centralized dashboard.

To ensure proper system design and development, Unified Modelling Language (UML) diagrams such as Use Case, Activity, Sequence, Class, Component, and Module Hierarchy diagrams are used to represent system architecture and interactions between system components. The system is implemented using PHP for backend services, HTML, CSS, JavaScript with Bootstrap for frontend development, and MySQL for database management.

By integrating artificial intelligence with modern web technologies, the PhishShield AI system provides an efficient solution for detecting phishing threats and improving cybersecurity awareness. The proposed system contributes toward building a safer digital environment by helping users identify and avoid malicious online content.

1.2 Existing System and Need for System

Currently, phishing detection is often dependent on traditional security methods such as static blacklists, manual verification, and browser warnings. These methods are limited because phishing websites frequently change domain names and structures to bypass detection mechanisms. Many users are unaware of phishing techniques and may unintentionally access malicious links or provide sensitive information through fraudulent emails and websites.

Additionally, existing systems often lack a centralized platform that allows users to easily analyze suspicious URLs or email content in real time. Most phishing detection tools operate independently and may not provide detailed threat analysis, risk scoring, or user-friendly interfaces for non-technical users.

There is a strong need for an intelligent phishing detection platform that allows users to quickly verify suspicious links and emails before interacting with them. Such a system should provide automated analysis, real-time threat detection, and easy-to-understand results to help users make informed decisions.

The proposed PhishShield AI system addresses this need by providing a secure web-based platform where users can scan URLs or email content, receive risk scores and threat levels, and access cybersecurity guidance through an AI chatbot. The system also enables administrators to monitor phishing trends, maintain threat databases, and generate reports for improved cybersecurity management.

1.3 Scope of System

The scope of the PhishShield AI system includes the development of a centralized platform that supports phishing detection and cybersecurity awareness through intelligent analysis and user interaction. The system covers the following functional areas:

- **User Interaction:** Users can register, log in, submit suspicious URLs or email content for analysis, view scan results, download reports, and interact with the AI chatbot. High-Level Interaction: Managing roles for the Admins, the Users, and Collectors.
- **Phishing Detection:** The system analyzes suspicious URLs and email content using feature extraction, blacklist verification, and AI-based classification techniques to detect potential phishing threats. Logistics: End-to-end tracking of pickup requests from initiation to completion.
- **Administrative Control:** Administrators manage users, maintain blacklist and whitelist domain databases, monitor scan activities, and analyze detection statistics through a dashboard.
- **Reporting and Analytics:** The system generates detailed reports and provides insights into phishing detection trends and system usage.
- **Security and Access Control:** Role-based access control ensures that system functionalities are accessed only by authorized users and administrators.

1.4 Operating Environment Hardware and Software:

Server-side requirement

Software Requirement	Hardware Requirement
Operating System: Windows 7 or above	Processor: Intel Core i3 or above
Front End: Add your Project front end here	RAM: 4GB or above
Back End: Add your Project back end here	HDD: 512 GB or above
Database: Add your Project database here	MySQL

Table No. 01: Represents the Server-side requirements of the PhishShield AI System.

Client-side requirement

Software Requirement	Hardware Requirement
Operating System: Windows 7 or above	Processor: Intel Core i3 or above
Web Browser: Chrome, Mozilla Firefox	RAM: 2GB or above
	HDD: 512 GB or above

Table No. 02: Represents the Client-side requirements of the PhishShield AI System.

1.5 Brief Description of Technology Used:

The PhishShield AI system is developed using a multi-tier web architecture that ensures scalability, security, and efficient processing of phishing detection requests.

- **UML Methodology:** Used for specifying, constructing, and documenting the system's development through Use Case, Activity, and Sequence diagrams.
- **PHP:** Used to develop the backend REST API services that handle authentication, scanning requests, and database operations**.**
- **HTML, CSS, JavaScript, Bootstrap:** Used to create the frontend user interface, providing a responsive and user-friendly web application.
- **MySQL Database:** Used for storing user information, scan results, feedback, blacklist domains, and system reports**.**
- **Artificial Intelligence Detection Module**: The AI module analyses URL features and email content patterns to classify potential phishing threats and generate risk scores.

CHAPTER 2: PROPOSED SYSTEM

2.1 Feasibility Study

A feasibility study is conducted to determine whether the proposed PhishShield AI – Intelligent Phishing Detection and Prevention System can be successfully implemented. The study evaluates the system from technical, operational, and economic perspectives.

- **Technical Feasibility:** The proposed system is technically feasible as it uses widely supported web technologies and frameworks. The backend is implemented using PHP, while the frontend is developed using HTML, CSS, JavaScript, and Bootstrap. The system stores data using a MySQL database, which provides reliable and efficient data management. The AI detection module analyzes suspicious URLs and email content to identify phishing threats. These technologies are widely used, well-supported, and compatible with modern computing environments.
- **Operational Feasibility:** The system is designed to be user-friendly and easy to operate. Users can register, log in, and submit suspicious URLs or email content for phishing analysis through a simple interface. Administrators can monitor system activity, manage phishing domain lists, and generate reports through an administrative dashboard. The system reduces manual analysis of phishing threats and provides automated detection, improving efficiency and user awareness.
- **Economic Feasibility:** The system is economically feasible because it relies on open-source technologies and standard web infrastructure. Development tools such as PHP, MySQL, and web technologies are freely available and do not require expensive licensing costs. The system also reduces potential financial losses caused by phishing attacks by helping users identify malicious links before interacting with them.

2.2 Objectives of the Proposed System

The main objectives of the PhishShield AI system are as follows:

1. To develop a web-based platform that allows users to analyze suspicious URLs and email content for phishing threats.
2. To implement an AI-based phishing detection mechanism that analyzes patterns and characteristics of malicious URLs and emails.
3. To provide real-time threat analysis and risk scoring, helping users determine whether online content is safe or malicious.

4. To create a secure and scalable system architecture using modern web technologies.
5. To allow administrators to monitor phishing detection activities and manage threat databases such as blacklist and whitelist domains.
6. To provide detailed reports and analytics that help understand phishing attack trends and system usage.
7. To improve cybersecurity awareness among users by providing guidance and educational support through the AI chatbot.

2.3 Users of the System

- **Admin:** The admin acts as the system manager responsible for maintaining and monitoring the platform. The admin can perform tasks such as managing users, maintaining blacklist and whitelist domain databases, analysing scan statistics, and generating security reports. The admin dashboard provides tools for monitoring system performance and managing phishing detection activities.
- **User:** The user represents individuals who interact with the system to detect phishing threats. Users can register, log in, and submit suspicious URLs or email content for phishing analysis. After analysis, users can view the risk score and threat level generated by the system. Users can also view previous scan reports, download reports, interact with the AI chatbot for cybersecurity advice, and submit feedback regarding system performance.

CHAPTER 3: ANALYSIS AND DESIGN

3.1 Entity Relationship Diagram (ERD)

The ERD represents the logical structure of the database used in the PhishShield AI system. The primary entities include:

- • **Admin:** Stores Admin ID, Username, and Password. Admin manages scan requests and dashboard activities.
- • **User:** Stores User ID, Name, Email, Password, Address, and Mobile Number. Users submit phishing scan requests and use system features.
- • **Scan Requests:** Contains Request ID, Scan Date, Risk Percentage, Threat Level, and Result. Requests are analysed by the AI Engine.
- • **AI Engine:** Detects phishing threats using Category ID, Category Name, and Confidence Score.
- • **Dashboard:** Displays statistics such as Total Users, Total Scans, and Total Threats.
- • **Chatbot:** Provides phishing-related assistance with Question and Response details.
- • **Feedback:** Stores user feedback using Feedback ID and Comment.

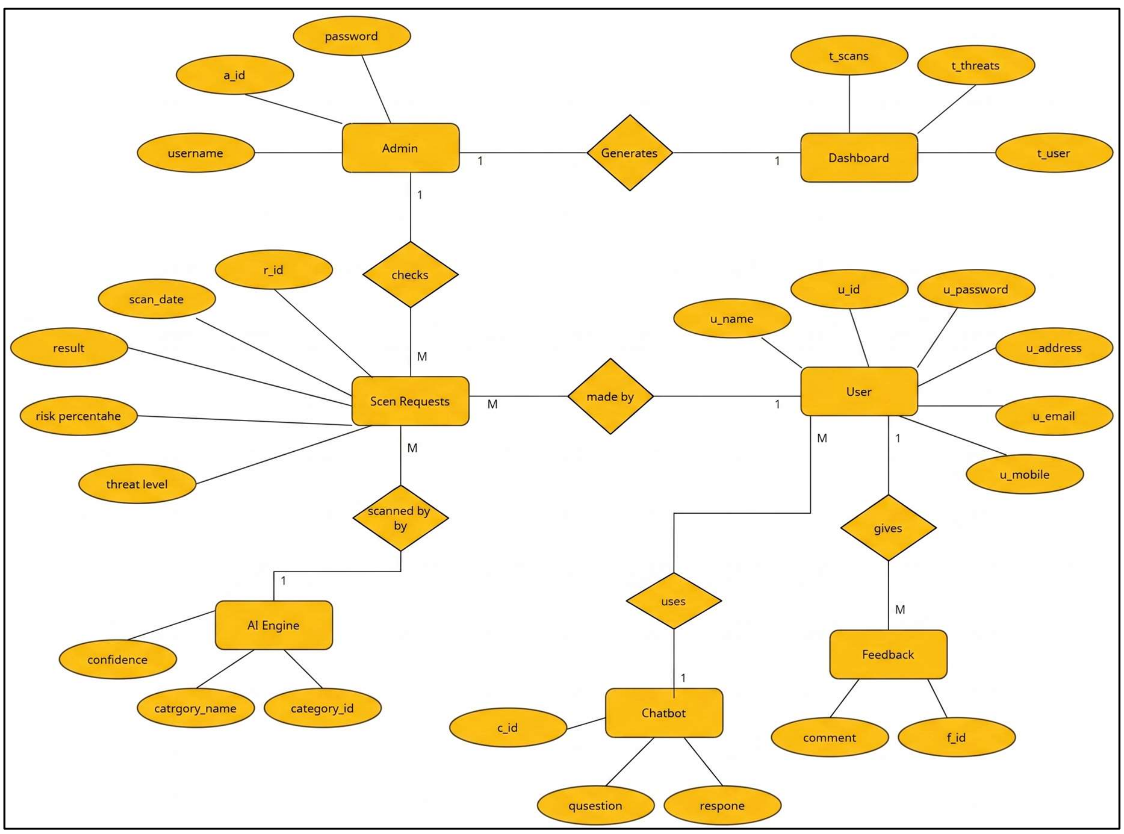

Figure No. 01: Represents the ER diagram of the PhishShield AI System

3.2 Class Diagram

The Class Diagram represents the static structure of the PhishShield AI System by showing the classes, their attributes, methods, and relationships within the system. It includes the following classes and functionalities:

- **Admin:** Manages users, blacklist/whitelist domains, and system analytics through the dashboard.
- **User:** Registers, logs in, performs phishing scans, views results, and submits feedback.
- **Dashboard:** Displays statistics such as total users, scans, threats, and analytics reports.
- **Scan:** Stores phishing scan details including URL, email content, risk score, threat level, and scan date.
- **AIDetectionEngine:** Analyses URLs and emails to detect phishing threats and calculate risk scores.
- **Blacklist:** Maintains records of malicious domains.
- **Whitelist:** Stores trusted domains approved by the admin.
- **Report:** Generates and downloads phishing scan reports.
- **Chatbot:** Provides automated responses to phishing-related queries.
- **Feedback:** Stores user comments and ratings for system improvement.

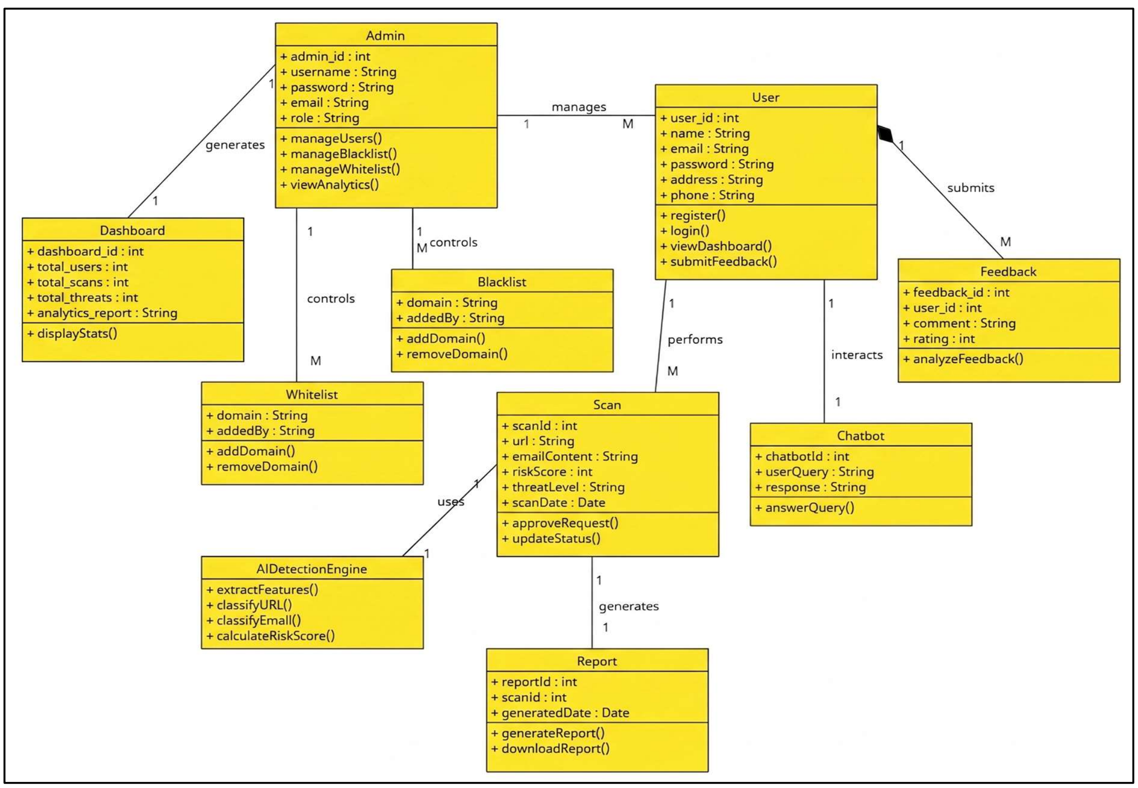

Figure No. 02: Represents the Class diagram of the PhishShield AI System

3.3 Use Case Diagrams

The Use Case Diagram illustrates the interactions between the actors (Admin, User, Collector) and the system functionalities.

Figure No. 03: Represents the Use case diagram of the PhishShield AI System

Use Case Name:	**PhishShield AI – Intelligent Phishing Detection System**
Actor:	Admin, User
Pre-Condition:	The user or admin must be registered and logged into the system.
Main Flow:	1. **Login Management –** Validate username and password, handle invalid login, and allow user authentication. 2. **User Management –** Users can register, login, and recover passwords if forgotten. 3. **Phishing Scan Management** – Users submit suspicious URLs or email content for analysis using the AI detection engine. 4. **Report Management –** Users can view scan reports, filter results, and download reports for analysis.

	5. **AI Chatbot Interaction –** Users can interact with the chatbot to get phishing help and cybersecurity advice. 6. **Feedback Management –** Users can submit feedback and view previous feedback related to system performance.
System Modules:	**Module1:** Login Module, **Module2:** User Registration Module, **Module3:** Phishing Scan Module, **Module4:** Report Module, **Module5:** AI Chatbot Module, **Module6:** Feedback Module.
Post-Condition:	The user will be logged into the system and able to perform activities such as scanning URLs or emails, viewing reports, interacting with the chatbot, and submitting feedback.

Table No. 03: Represents the Use Case Description of the PhishShield AI System.

3.4 Activity Diagram

The Activity Diagram depicts the dynamic flow of the PhishShield AI system and shows how different operations are performed by users and administrators.

System Flow:

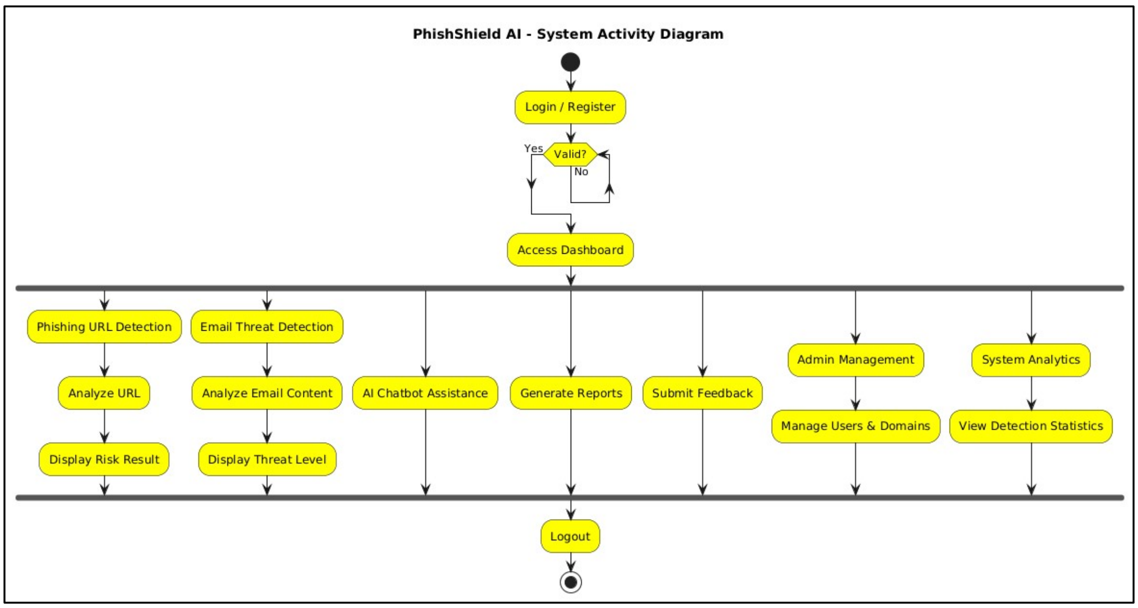

Figure No. 04: Represents the Admin Activity diagram of the PhishShield AI System

The Fig. 4 refers to the System Activity Diagram of the PhishShield AI System, illustrating the overall workflow of the system. It shows the process starting from user login and validation, followed by access to the dashboard. The diagram includes activities such as phishing URL detection, email threat detection, AI chatbot assistance, report generation, feedback submission, admin management, and system analytics. It also represents the analysis process for URLs and emails, display of threat results, management of users and domains, and monitoring of detection statistics. The activity flow ends with the logout process.

Admin Flow:

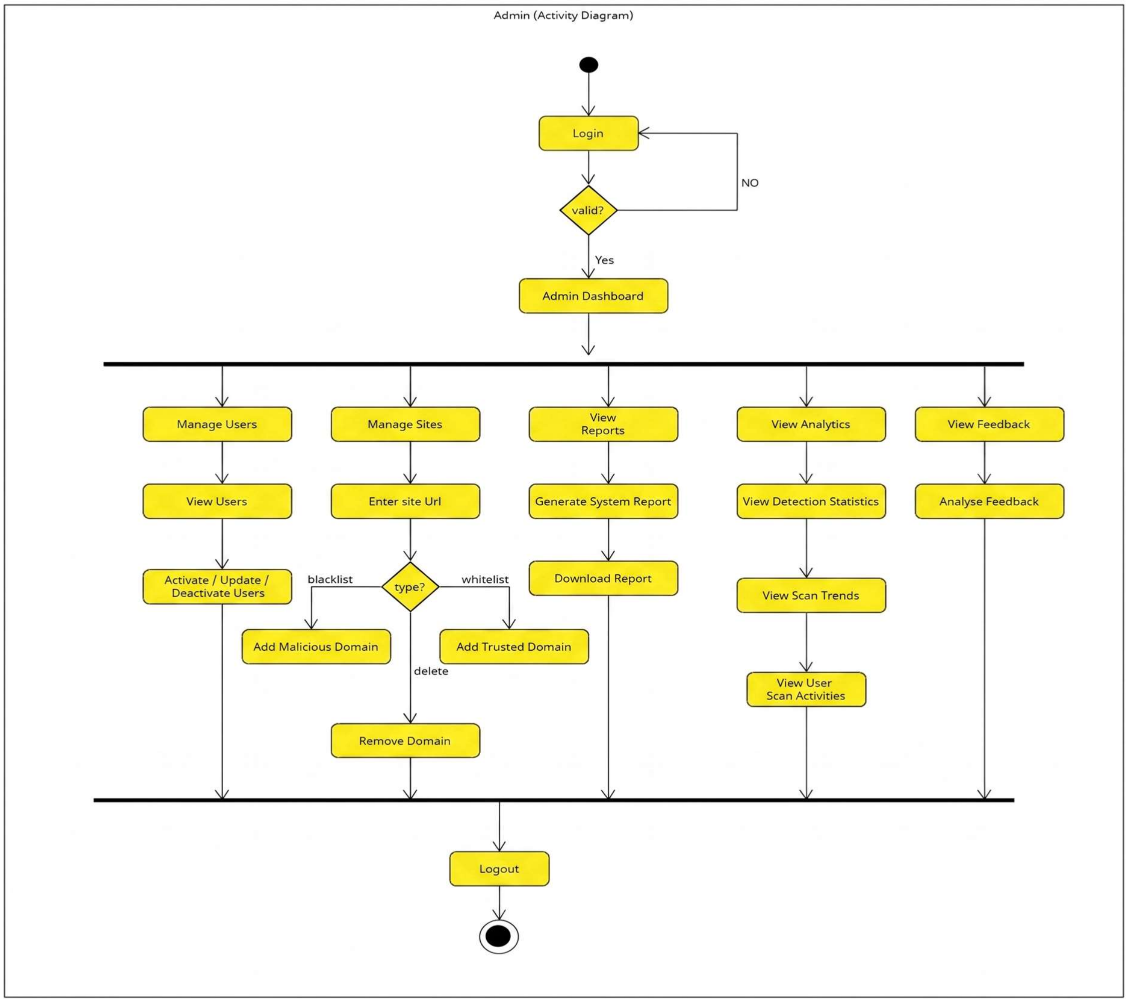

Figure No. 05: Represents the Admin Activity diagram of the PhishShield AI System

The Figure No. 05 refers to the Admin Activity Diagram of the PhishShield AI System, illustrating the workflow of admin operations within the system. It shows the process beginning with admin login and validation, followed by access to the admin dashboard. The diagram includes activities such as managing users, managing malicious and trusted domains, generating and downloading reports, viewing analytics and detection statistics, monitoring user scan activities, analyzing feedback, and maintaining blacklist and whitelist records. The activity flow ends with the logout process.

User Flow: Starts with login. If the user is not registered, they complete registration. After successful login, the user accesses the dashboard where they can scan suspicious URLs or email content, view reports, interact with the AI chatbot, and submit feedback. The phishing scan process involves entering suspicious data, analyzing it using the AI detection engine, calculating the risk score, and displaying the threat level. The process ends with the user logging out.

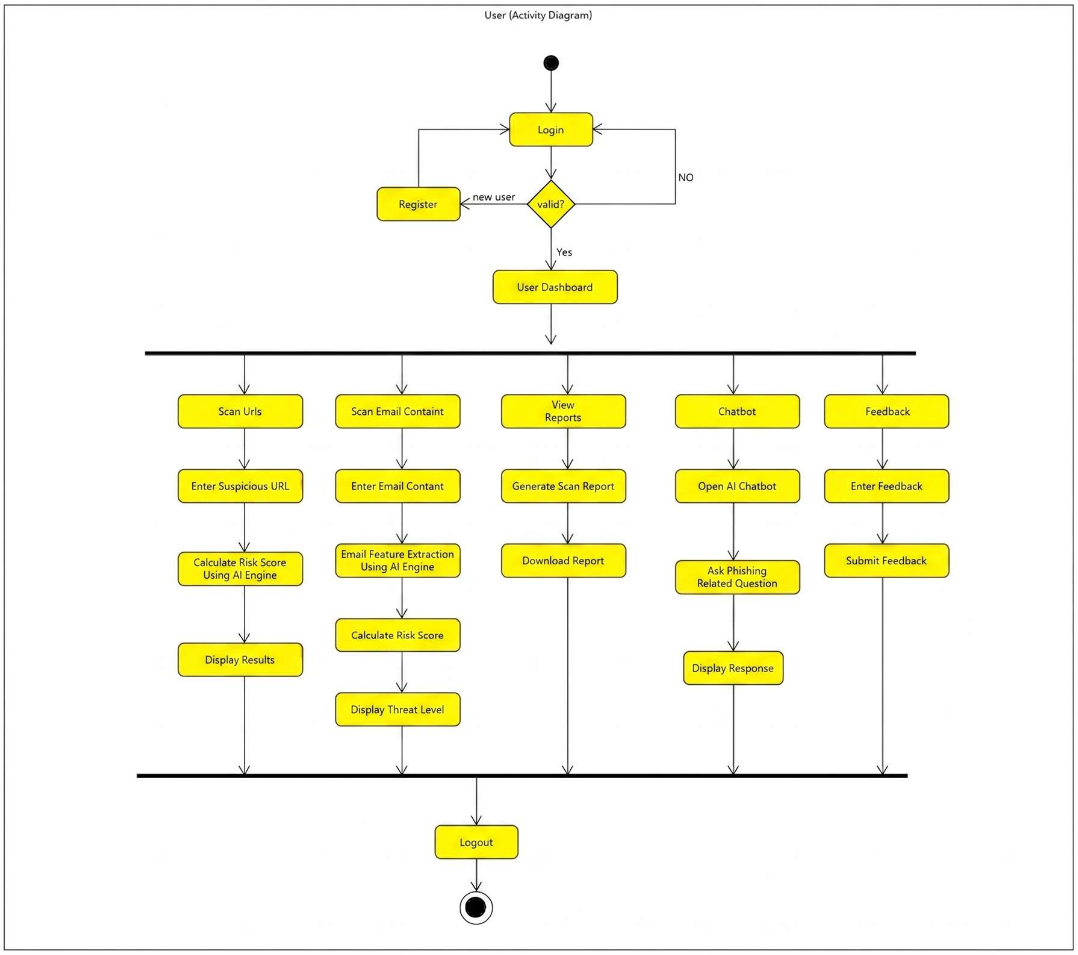

Figure No. 06: Represents the User Activity diagram of the PhishShield AI System

Figure No. 06: Represents the User Activity Diagram of the PhishShield AI System, illustrating the workflow of user operations such as registration, login validation, phishing URL and email scanning, AI-based risk analysis, report generation, chatbot interaction, feedback submission, and logout process.

3.5 Sequence Diagram

System Sequence: The Sequence Diagrams illustrate the chronological interactions between system components and actors in the PhishShield AI system. These diagrams demonstrate how messages are exchanged between the user interface, system services, AI detection engine, and database to perform phishing detection and management operations.

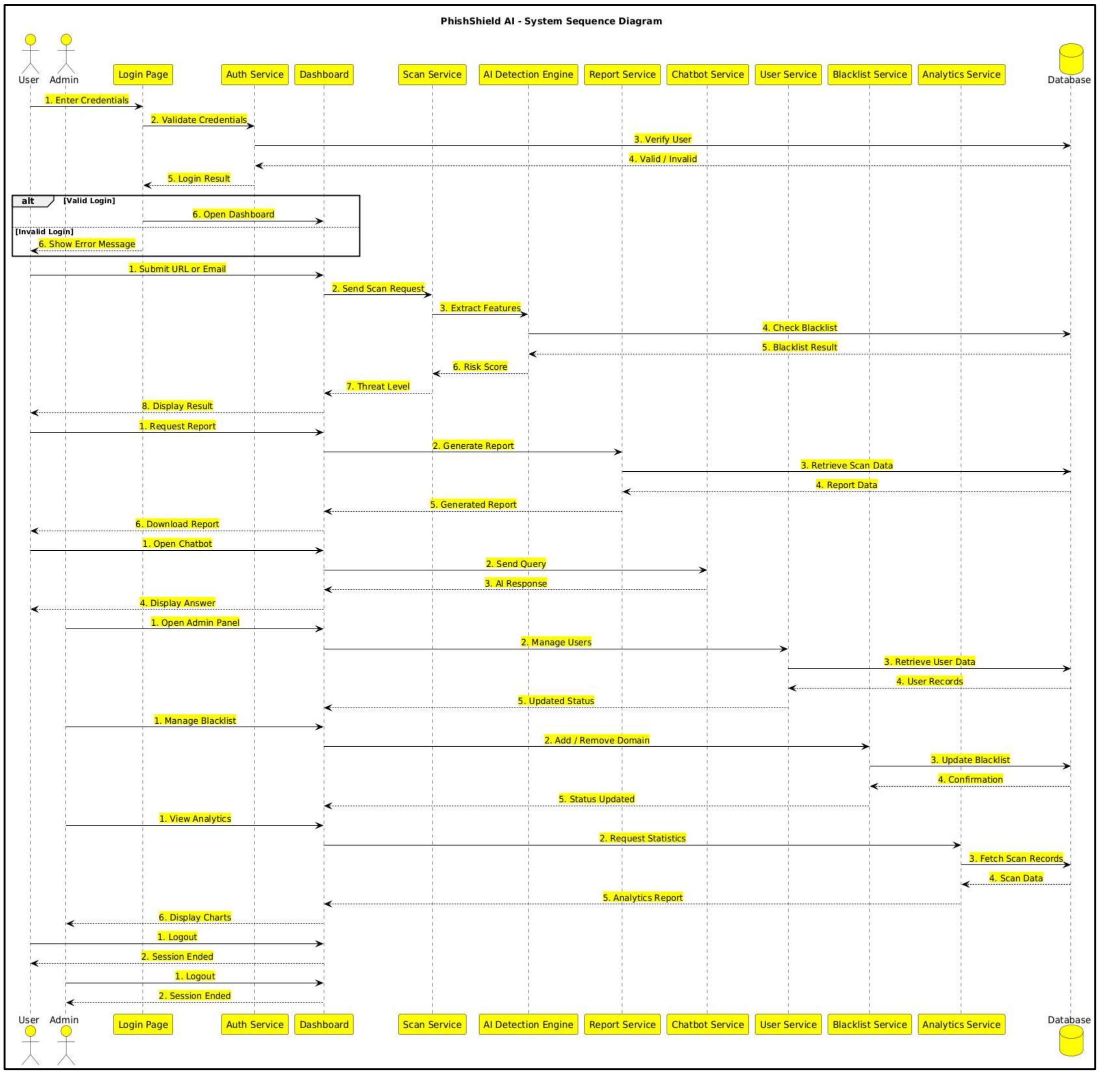

Figure No. 07: Represents the System Sequence diagram of the PhishShield AI System

Figure No. 07: Represents the System Sequence Diagram of the PhishShield AI System, illustrating the interaction between users, admin, system services, and database during processes such as login authentication, phishing URL and email scanning, AI-based threat detection, report generation, chatbot communication, blacklist management, analytics monitoring, and logout operations.

User Sequence:

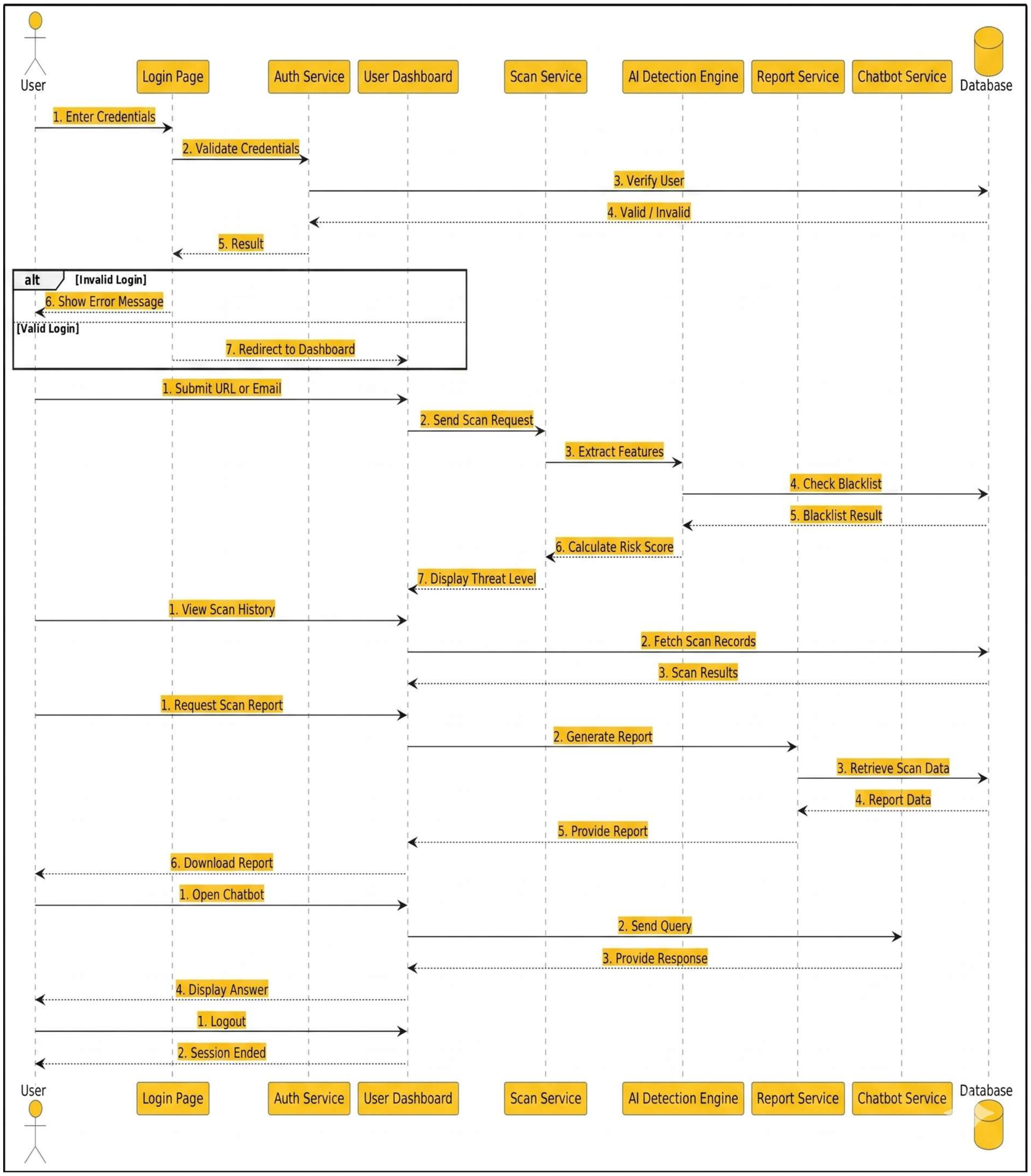

Figure No. 08: Represents the Admin Activity diagram of the PhishShield AI System

Figure No. 08: Represents the User Sequence Diagram of the PhishShield AI System, illustrating the sequence of interactions between the user and system components during login authentication, phishing URL or email scanning, AI-based risk analysis, viewing scan history, report generation, chatbot interaction, and logout process.

Admin Sequence Diagram:

Details the interaction when an Admin validates credentials, manages the user/collector database, and processes pickup approvals or rejections based on incoming requests.

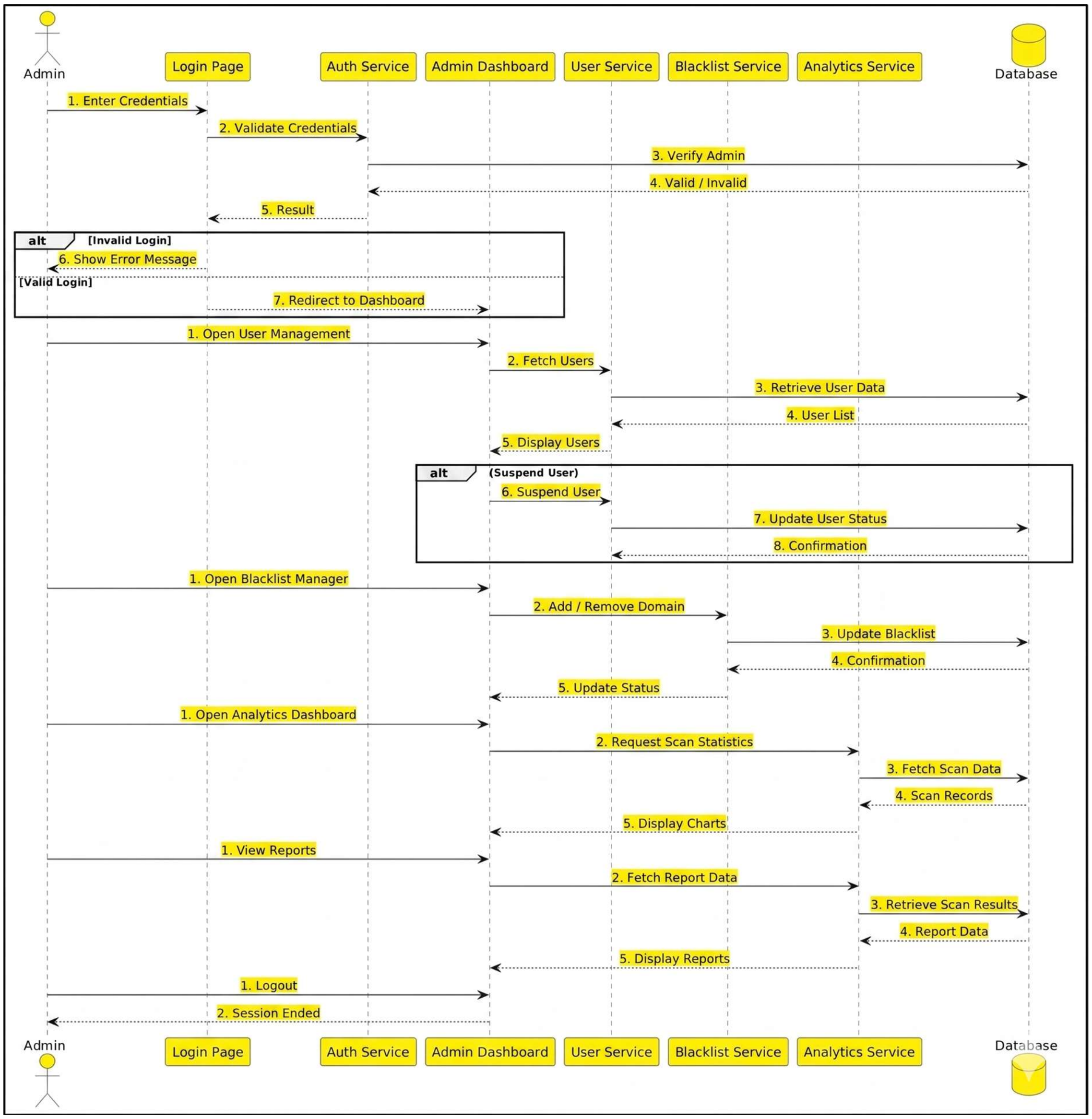

Figure No. 09: Represents the Admin Activity diagram of the PhishShield AI System

Figure No. 09: Represents the Admin Sequence Diagram of the PhishShield AI System, illustrating the sequence of interactions between the admin and system components during login authentication, user management, blacklist management, analytics monitoring, report viewing, and logout process.

3.6 Component Diagram

The Component Diagram illustrates the high-level structural composition of the PhishShield AI System, highlighting the modular architecture and the dependencies between its core subsystems. It shows how components such as the User Interface, Chatbot, Scan Request Module, Report Module, Security Services (Encryption and Access Control), Persistence Layer, and Database interact to support the overall functionality of the system.

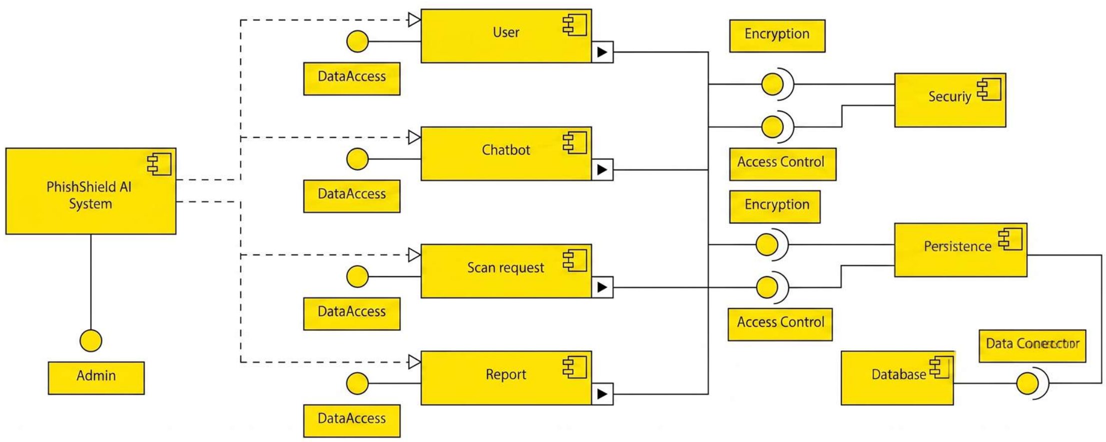

Figure No. 10: Represents the Component diagram of the PhishShield AI System

Key Components and Functionalities:

- **User & Chatbot Modules:** These modules serve as the primary interaction layer between the users and the PhishShield AI System. The User module allows users to register, log in, submit suspicious URLs or emails, and access reports. The Chatbot module provides phishing-related guidance, awareness information, and automated responses to user queries. Both modules communicate with backend services through secure interfaces.
- **Scan Request Module:** This component processes phishing scan requests submitted by users. It receives suspicious links, messages, or email content and forwards them to the AI analysis services for phishing detection and risk classification.
- **Report Module:** Responsible for generating detailed phishing detection reports that include scan outcomes, threat levels, and analysis results. These reports help users and administrators monitor phishing activities and system performance.
- **Data Access Layer:** Acts as an intermediate layer between the application modules and backend services. It manages data communication and ensures efficient interaction between system components and the database.

- **Security Interface:** Ensures system protection and secure communication between components. It includes:
    - **Access Control:** Restricts unauthorized access and allows only authenticated users or administrators to access sensitive features and system data.
    - **Encryption:** Protects transmitted data by converting it into a secure format to prevent unauthorized access or manipulation.
- **Persistence & Data Connector:** These components manage the storage and retrieval of system data. The Persistence layer handles long-term data storage, while the Data Connector establishes communication with the database system.
- **Database:** Serves as the central repository of the PhishShield AI System. It stores user profiles, phishing scan requests, AI detection results, generated reports, feedback records, and other system-related information.

3.7 Module and Hierarchy Diagram

The Module and Hierarchy Diagram (based on the Package Diagram) illustrate the logical grouping of functionalities into cohesive modules and the hierarchical relationships between them. This view is important for understanding the modular decomposition of the PhishShield AI system, where related functionalities are organized into independent yet interconnected packages.

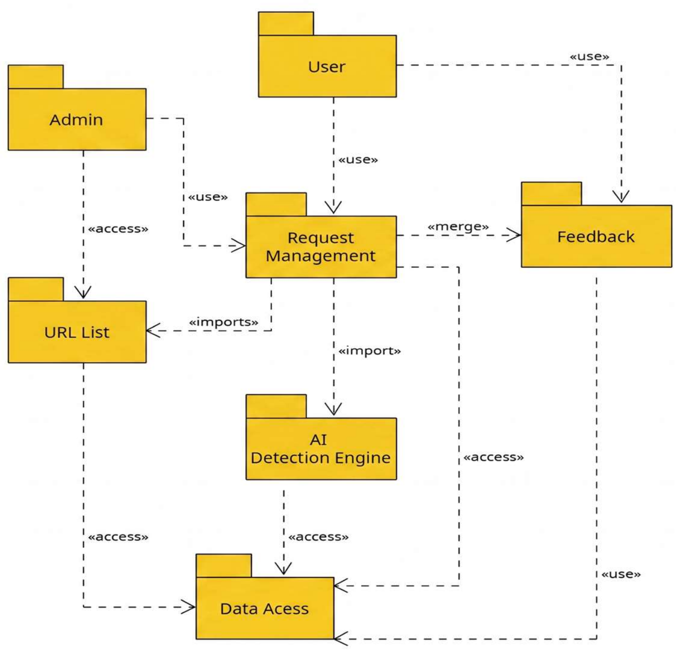

Figure No. 11: Represents the Package diagram of the PhishShield AI System

Technical Analysis of Modules:

- **Presentation Tier Modules:** The Module and Hierarchy Diagram (based on the Package Diagram) illustrate the logical grouping of functionalities into cohesive modules and the hierarchical relationships between them. This view is important for understanding the modular decomposition of the PhishShield AI system, where related functionalities are organized into independent yet interconnected packages. The modular structure improves maintainability, scalability, and clarity in system design by separating responsibilities across different layers of the application.
- **Service Tier Modules:**
  - **Scan Request Module:** Handles the core logic for submitting and processing phishing scan requests. It manages user inputs such as suspicious links, emails, or messages and forwards them to the AI analysis engine for classification.
  - **AI Engine Module:** A specialized module responsible for analysing the submitted data using phishing detection algorithms and classification models. It generates intelligent results that determine whether the content is malicious or safe.
  - **Feedback Module:** Collects feedback from users regarding detection results and system responses. This information helps improve system performance and supports future learning and refinement of the AI detection mechanisms.
- **Support & Infrastructure Modules:**
  - **Security Module:** Manages security mechanisms such as access control and encryption to protect sensitive system data and ensure that only authorized users can perform administrative or critical operations.
  - **Data Access Module:** Acts as a foundational infrastructure component that provides a standardized interface for communication between the service modules and the database. It handles data storage, retrieval, and interaction with the persistence layer, ensuring reliable management of user data, scan results, and system logs.

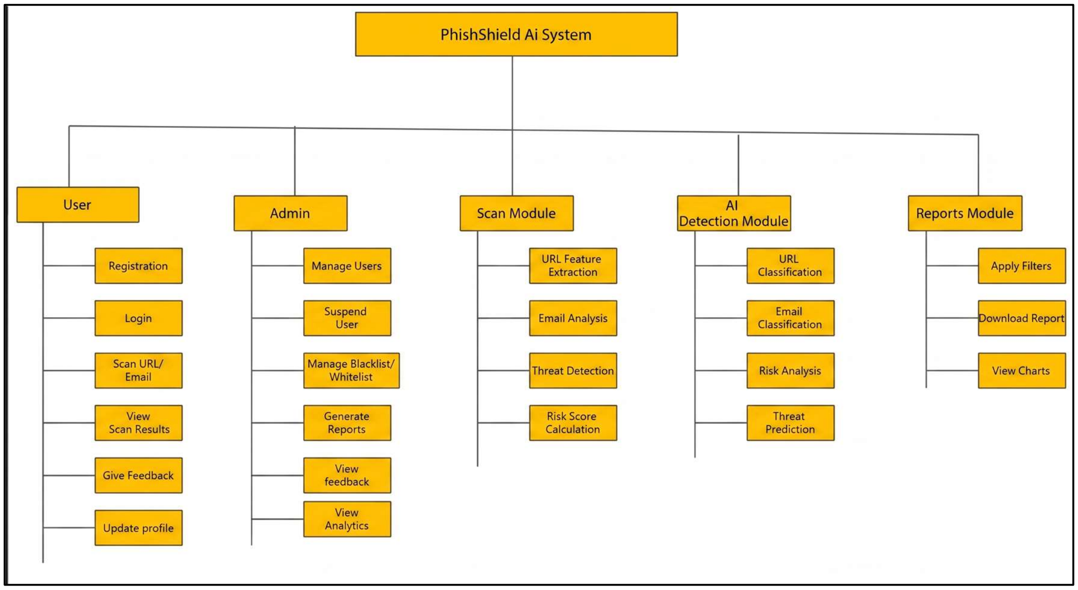

Figure No. 12: Represents the Hierarchy diagram of the PhishShield AI System.

Hierarchical Flow: The system follows a top-down hierarchical structure where the actor modules (User and Admin) initiate system operations that interact with service modules such as Scan Module, AI Detection Module, and Reports Module. These service modules further depend on underlying infrastructure components to process requests and securely store data in the central database.

3.8 Table Design

The database for the E-Waste Management System is designed to ensure data integrity and efficient retrieval. The primary tables are:

1. **Admin:** Stores administrative credentials and profile details used for system monitoring and management.
2. **User:** Stores registration details of users who submit suspicious URLs or emails for phishing analysis.
3. **Scan Requests:** Central transaction table that records all phishing scan submissions made by users, including URLs or email content.
4. **Detection Results:** Stores the results generated by the AI detection module, including classification results, risk scores, and threat predictions.
5. **Feedback:** Stores user feedback regarding scan results and system responses to help improve the detection process.

3.9 Data Dictionary

The data dictionary provides a detailed description of the tables, including data types and constraints.

Admin:

Field Name	Data Type	Size	Constraint	Description
admin_id	INT	11	Primary Key	Unique identification for each admin
username	VARCHAR	50	Unique	Login username for admin
password	VARCHAR	255	Not Null	Encrypted password
email	VARCHAR	50	Unique	Admin email address
last_login	DATETIME	20	Not Null	Timestamp of the last login session

User:

Field Name	Data Type	Size	Constraint	Description
user_id	INT	11	Primary Key	Unique identification for each user
name	VARCHAR	50	Not Null	Full name of the user
email	VARCHAR	50	Unique	Email address for login
password	VARCHAR	255	Not Null	Encrypted password
address	TEXT	200	Not Null	Residential address for pickups
contact_no	BIGINT	10	Not Null	10-digit mobile number

Scan Requests:

Field Name	Data Type	Size	Constraint	Description
request_id	INT	11	Primary Key	Unique scan request ID
user_id	INT	11	Foreign Key	Reference to User table
scan_type	VARCHAR	20	Not Null	Type of scan (URL / Email)
content	TEXT	–	Not Null	URL or email content submitted
status	VARCHAR	20	Default Pending'	Current scan processing status
request_date	DATETIME	–	–	Date when the scan request was submitted

Detection Results:

Field Name	Data Type	Size	Constraint	Description
result_id	INT	11	Primary Key	Unique detection results ID
request_id	INT	11	Foreign Key	Reference to Scan Requests table
classification	VARCHAR	20	Not Null	Result (Phishing / Safe)
risk_score	FLOAT	–	–	Risk probability score
threat_level	VARCHAR	20	–	Level of threat detected
analysis_date	DATETIME	–	–	Date of analysis

Feedback:

Field Name	Data Type	Size	Constraint	Description
feedback_id	INT	11	Primary Key	Unique feedback ID
request_id	INT	11	Foreign Key	Reference to Scan Requests table
rating	INT	1	Check (1–5)	User satisfaction rating
comments	TEXT	–	–	User feedback comments
feedback_date	DATETIME	–	–	Date feedback was submitted

3.10 Sample Input and Output Screens

Home Page:

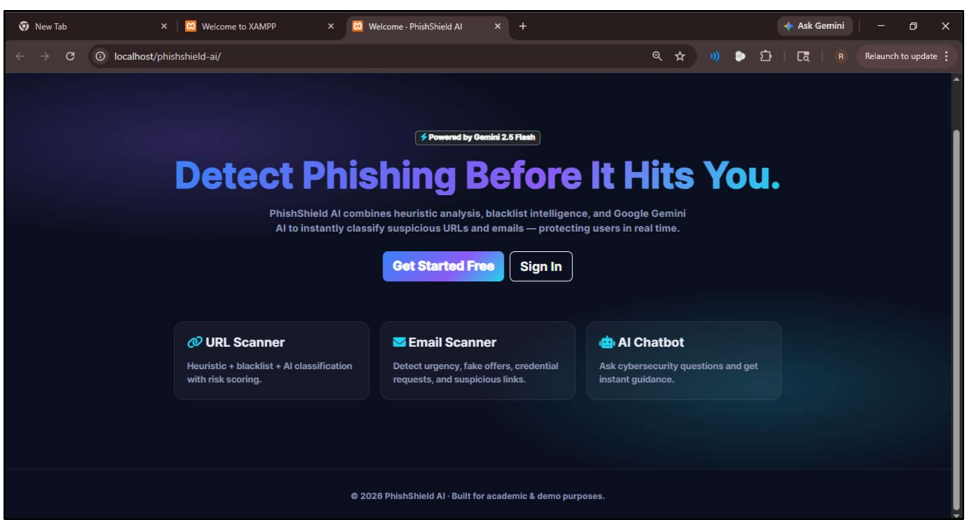

Figure No. 13: Represents Home Page of PhishShield AI System

The Figure No. 13 represents the Home Page of the PhishShield AI System. It provides an overview of the system's main features, including URL scanning, email scanning, and AI chatbot assistance for phishing detection. The page allows users to access the Sign In and Get

Started options for authentication and registration. It also highlights the use of AI-powered phishing analysis to classify suspicious URLs and emails in real time for improved cybersecurity awareness and protection.

Registration Page:

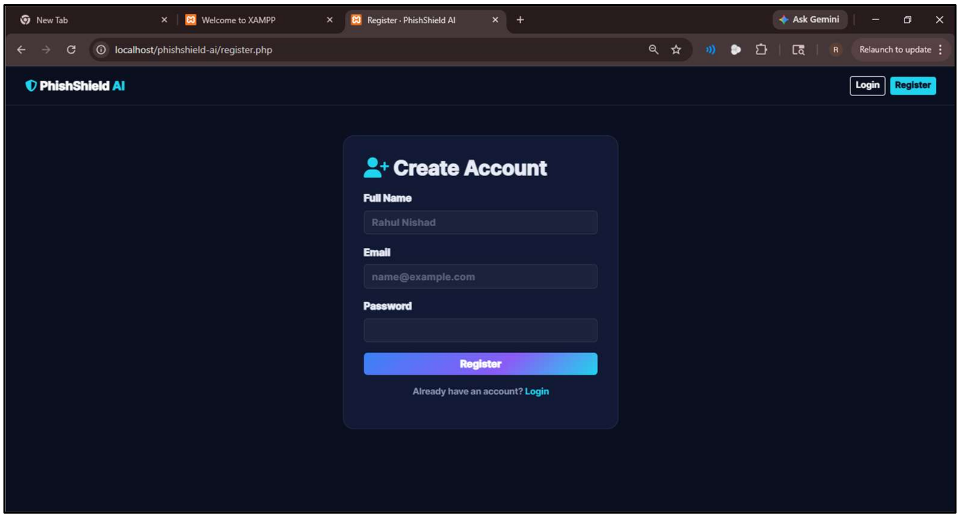

Figure No. 14: Represents Registration Page of PhishShield AI System

The figure no. 14 shows the user registration page of the PhishShield AI System, where users can create an account using basic details like name, email, and password. The page includes password validation to ensure secure credentials. It provides a simple and user-friendly interface for accessing phishing detection services.

Login Page:

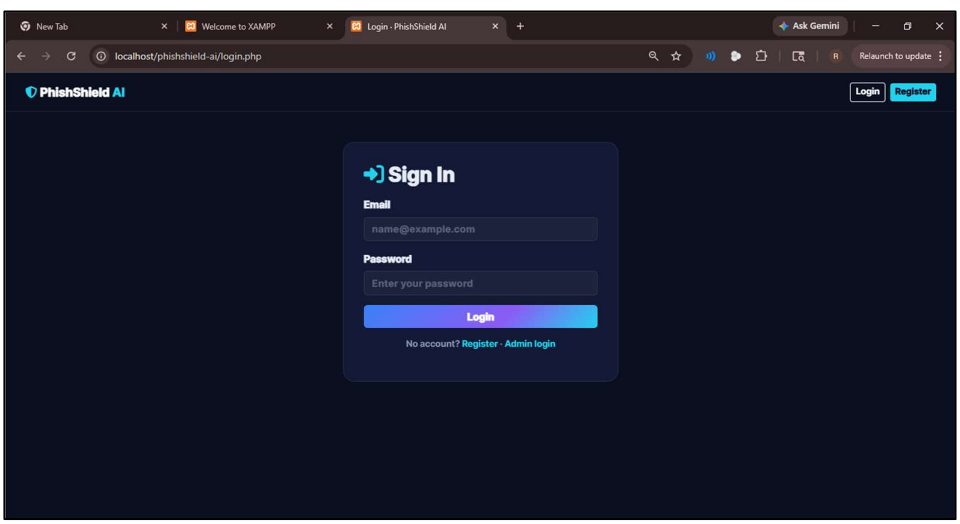

Figure No. 15: Represents Login Page of PhishShield AI System

The figure no. 15 represents the login page of the PhishShield AI System. It allows registered users to securely access their accounts using their email address and password. The page also includes features such as Remember Me and Forgot Password options for improved usability, along with a link for new users to create an account.

Admin Dashboard:

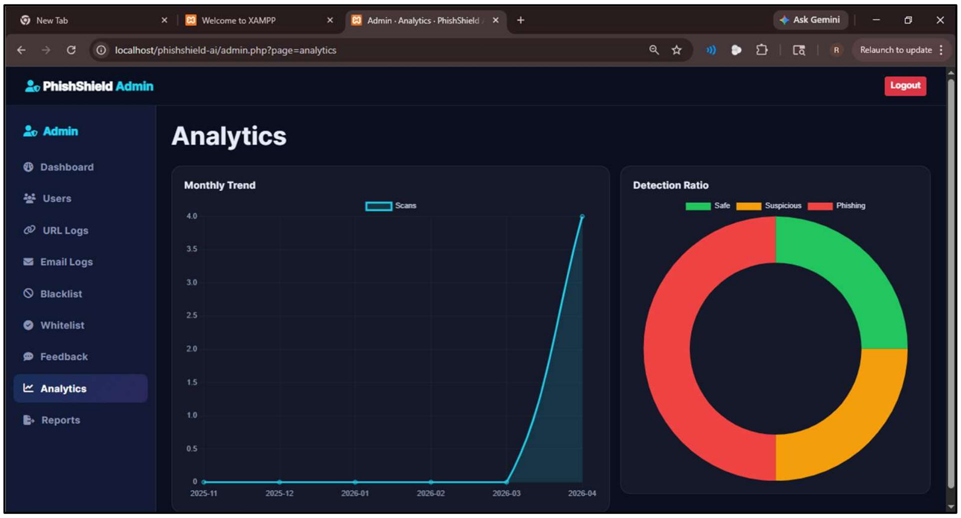

Figure No. 16: Admin Dashboard of PhishShield AI System.

The figure no. 16 illustrates the Admin Dashboard of the PhishShield AI System, displaying key metrics such as total scans, active users, and detection rate. It provides insights into phishing trends and system performance. The dashboard helps administrators monitor activities and manage the system efficiently

User Dashboard 1:

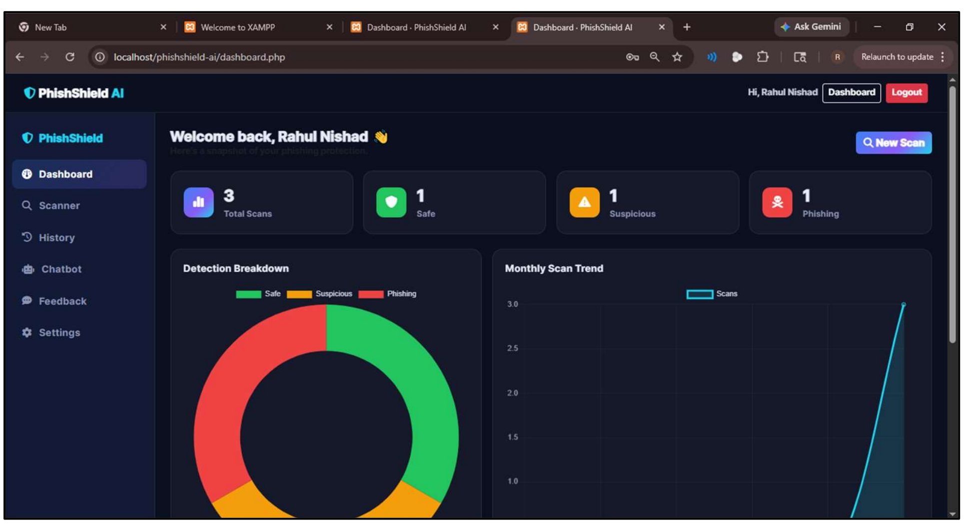

Figure No. 17: User Dashboard PhishShield AI System.

The figure no. 17 shows the User Dashboard of the PhishShield AI System, allowing users to scan URLs or emails and view detection results. It provides an overview of scan statistics, recent activity, and threat distribution. The dashboard also enables users to access scan history and monitor phishing threats efficiently.

User Dashboard 2 (Feedback):

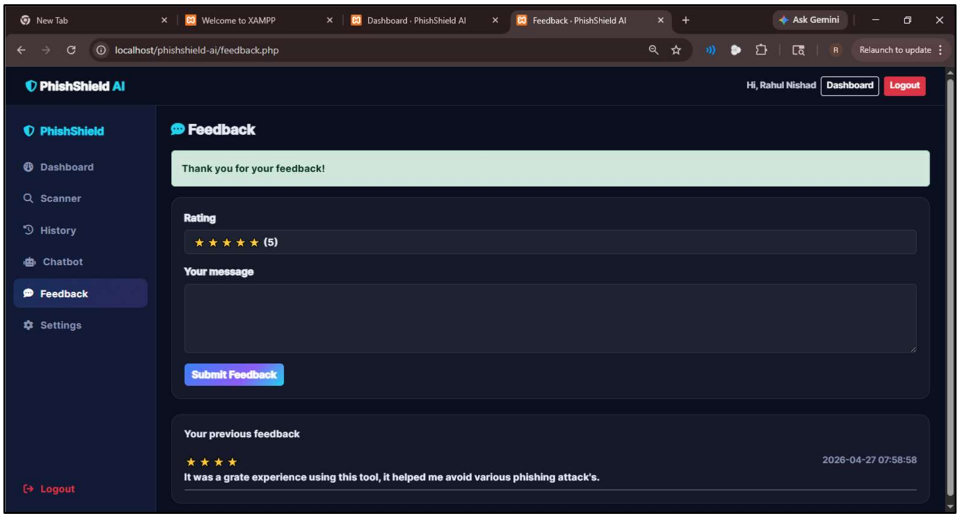

Figure No. 18: User Dashboard (Feedback) of PhishShield AI System.

The figure no. 18 represents the Feedback section of the PhishShield AI System, where users can share their experience after using the platform. It allows users to rate the system and submit feedback or report issues through comments. This module helps improve system performance and enhances phishing detection accuracy based on user input.

User Dashboard 2 (URL Scanner):

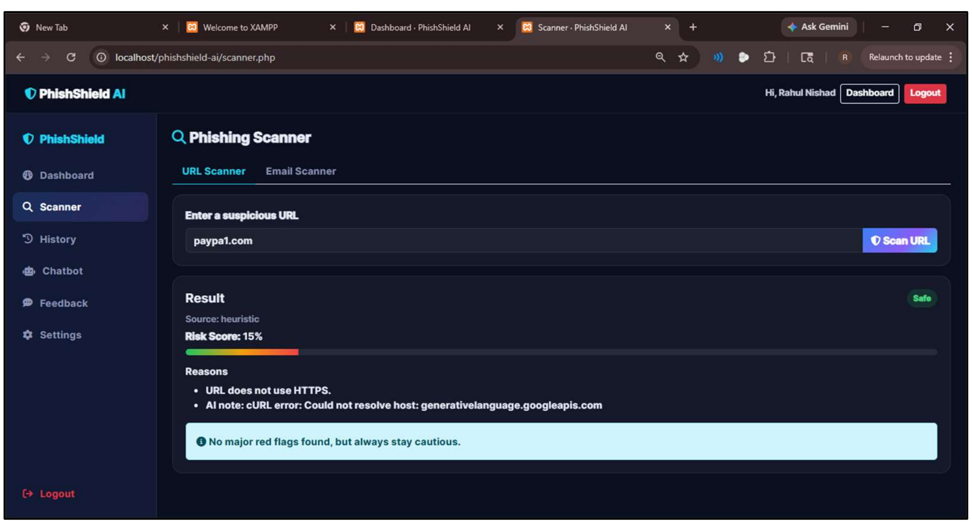

Figure No. 19: URL Scanner Module of PhishShield AI System.

The figure no. 19 shows the URL Scanner module of the PhishShield AI System, where users can analyze URLs for phishing threats. It displays the risk level, risk score, and detection methods used for classification. This module helps users identify malicious links and enhances overall security awareness.

CHAPTER 4: COADING Sample Code

Report Code:

```php
<?php
/**
 * PhishShield AI - CSV Reports Export (admin only)
 */
require_once __DIR__ . '/config/auth.php';
require_admin();

$type = $_GET['type'] ?? '';
$allowed = ['url_scans','email_scans','users','feedback'];
if (!in_array($type, $allowed)) { http_response_code(400); exit('Invalid report type.'); }

$file = "phishshield_{$type}_" . date('Ymd_His') . ".csv";
header("Content-Type: text/csv; charset=utf-8");
header("Content-Disposition: attachment; filename=\"$file\"");

$out = fopen('php://output', 'w');

switch ($type) {
    case 'url_scans':
        fputcsv($out, ['ID','User ID','URL','Result','Risk','Source','Reasons','Recommendation','Date']);
        foreach ($pdo->query("SELECT * FROM url_scans ORDER BY id") as $r)
            fputcsv($out, [$r['id'],$r['user_id'],$r['url'],$r['result'],$r['risk_score'],$r['source'],$r['reasons'],$r['recommendation'],$r['created_at']]);
        break;
    case 'email_scans':
        fputcsv($out, ['ID','User ID','Content','Result','Risk','Reasons','Date']);
        foreach ($pdo->query("SELECT * FROM email_scans ORDER BY id") as $r)
            fputcsv($out, [$r['id'],$r['user_id'],$r['content'],$r['result'],$r['risk_score'],$r['reasons'],$r['created_at']]);
        break;
    case 'users':
        fputcsv($out, ['ID','Name','Email','Blocked','Created']);
        foreach ($pdo->query("SELECT id,name,email,is_blocked,created_at FROM users ORDER BY id") as $r)
            fputcsv($out, [$r['id'],$r['name'],$r['email'],$r['is_blocked'],$r['created_at']]);
        break;
    case 'feedback':
        fputcsv($out, ['ID','User ID','Rating','Message','Date']);
        foreach ($pdo->query("SELECT * FROM feedback ORDER BY id") as $r)
            fputcsv($out, [$r['id'],$r['user_id'],$r['rating'],$r['message'],$r['created_at']]);
        break;
}
fclose($out);
```

Figure No. 20: Represents the implementation code of the Report Export Module of the PhishShield AI System.

Report Output Page:

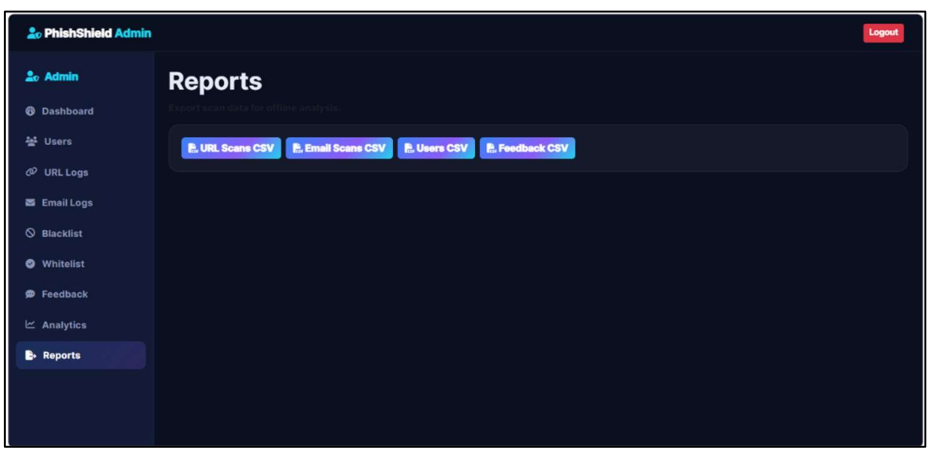

Figure No. 21: Represents the Reports Page of the PhishShield AI System.

Chatbot Module Code:

```
<?php
require_once __DIR__ . '/config/auth.php';
require_once __DIR__ . '/config/functions.php';
require_once __DIR__ . '/config/gemini.php';
require_login();
$user = current_user($pdo);

if ($_SERVER['REQUEST_METHOD'] === 'POST' && isset($_POST['msg'])) {
    header('Content-Type: application/json');
    $msg = trim($_POST['msg']);
    if ($msg === '') { echo json_encode(['reply' => '...']); exit; }
    $stmt = $pdo->prepare("INSERT INTO chat_history (user_id,role,message) VALUES (?,?,?)");
    $stmt->execute([$user['id'],'user',$msg]);
    $reply = chatBotReply($msg);
    $stmt->execute([$user['id'],'bot',$reply]);
    echo json_encode(['reply' => $reply]);
    exit;
}

$hist = $pdo->prepare("SELECT role, message FROM chat_history WHERE user_id=? ORDER BY id ASC LIMIT 50");
$hist->execute([$user['id']]);
$hist = $hist->fetchAll();

$pageTitle = 'AI Chatbot';
include __DIR__ . '/includes/header.php';
include __DIR__ . '/includes/navbar.php';
?>
<div class="ps-app">
  <?php include __DIR__ . '/includes/sidebar.php'; ?>
  <main class="ps-main fade-in">
    <div class="ps-page-head"><h1><i class="fa-solid fa-robot text-cyan"></i> Cybersecurity Chatbot</h1></div>

    <div class="ps-card chat-box">
      <div class="chat-log" id="chatLog">
        <?php if (!$hist): ?>
          <div class="chat-msg bot">👋 Hi <?= e($user['name']) ?>! I'm PhishShield AI. Ask me anything about phishing, scams or staying safe online.</div>
        <?php else: foreach ($hist as $h): ?>
          <div class="chat-msg <?= $h['role']==='user'?'user':'bot' ?>"><?= e($h['message']) ?></div>
        <?php endforeach; endif; ?>
      </div>
      <form id="chatForm" class="d-flex gap-2 mt-3">
        <input id="chatInput" class="form-control" placeholder="Ask something like: How do I spot a phishing email?" required>
        <button class="btn btn-grad"><i class="fa-solid fa-paper-plane"></i></button>
      </form>
    </div>
  </main>
</div>

<script>
const log  = document.getElementById('chatLog');
const form = document.getElementById('chatForm');
const inp  = document.getElementById('chatInput');

function addMsg(role, text) {
  const d = document.createElement('div');
  d.className = 'chat-msg ' + role;
  d.textContent = text;
  log.appendChild(d);
  log.scrollTop = log.scrollHeight;
  return d;
}
log.scrollTop = log.scrollHeight;

form.addEventListener('submit', async (e) => {
  e.preventDefault();
  const msg = inp.value.trim();
  if (!msg) return;
  addMsg('user', msg);
  inp.value=''; inp.disabled=true;
  const thinking = addMsg('bot','Thinking…');
  try {
    const fd = new FormData(); fd.append('msg', msg);
    const r = await fetch('chatbot.php', { method:'POST', body: fd });
    const j = await r.json();
    thinking.textContent = j.reply || '(no reply)';
  } catch (err) {
    thinking.textContent = 'Error contacting AI.';
  }
  inp.disabled=false; inp.focus();
});
</script>
<?php include __DIR__ . '/includes/footer.php'; ?>
```

Figure No. 22: Represents the implementation code of the AI Chatbot Module of the PhishShield AI System.

Chatbot Module Output Page:

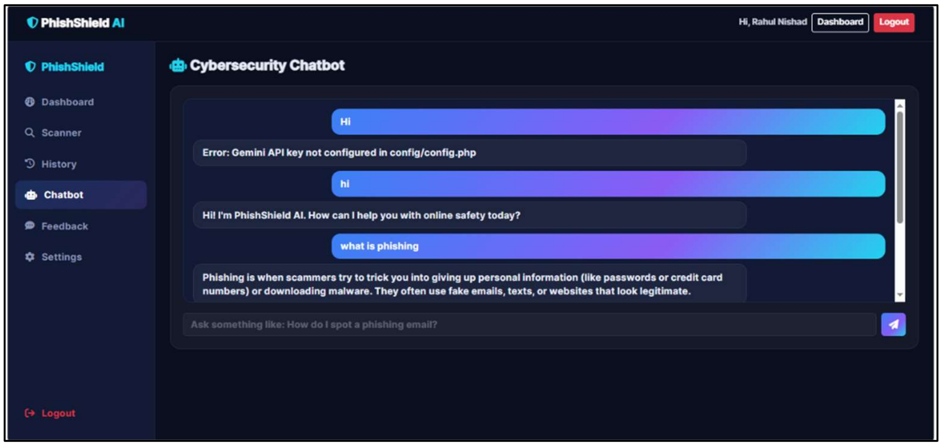

Figure No. 23: Represents the AI Chatbot Page of the PhishShield AI System.

The Figure No. 23 represents the AI Chatbot Page of the PhishShield AI System. It allows users to interact with the cybersecurity chatbot by asking phishing-related questions and receiving automated responses. The chatbot helps users understand phishing attacks, suspicious links, online scams, and cybersecurity safety measures through real-time AI-based guidance.

CHAPTER 5: LIMITATIONS OF SYSTEM

The E-Waste Management System, while robust, has several limitations that must be addressed for future scalability:

1. **Mandatory Internet Connectivity:** As a web-based application utilizing AI Classification Servers and centralized databases, an active internet connection is essential. The system cannot perform AI analysis or sync pickup data in offline mode.
2. **AI Image Quality Sensitivity:** The accuracy of the AI Engine's image classification is highly dependent on the quality of user-uploaded photos. Low-resolution, blurry, or poorly lit images can lead to misclassification of electronic components.
3. **Physical Inspection Constraints:** The system facilitates digital requests but cannot verify the internal hardware condition or functional state of the e-waste until the collector performs a physical on-site inspection.
4. **Geographical Boundaries:** Initial deployment is limited to specific operational area codes. Users residing in remote zones or outside the serviced pin codes may find the platform inaccessible for pickup scheduling.

5. **Browser & Hardware Dependencies:** Performance on the client side is subject to the user's hardware specifications and web browser version. Legacy browsers may experience rendering issues with modern React.js components.

CHAPTER 6: PROPOSED ENHANCEMENTS

To improve the efficiency and reach of the PhishShield AI System, the following enhancements are proposed for future development:

1. **Dependence on Dataset Quality:** The accuracy of the system heavily depends on the quality and diversity of the training dataset. If the dataset does not include newly emerging phishing patterns, the model may fail to detect them effectively.
2. **Difficulty in Detecting Zero-Day Phishing Attacks:** The system may struggle to identify new or previously unseen (zero-day) phishing websites, as they may not match known patterns or features used during training.
3. **False Positives and False Negatives:** There is a possibility that legitimate websites may be flagged as phishing (false positives) or malicious websites may be marked as safe (false negatives), which can affect user trust.
4. **Limited Real-Time URL Analysis**: If the system relies on static features or offline models, it may not fully analyze real-time factors such as dynamic page content, JavaScript behavior, or live redirects.
5. **Lack of Deep Content Inspection:** The current system may focus mainly on URL-based features (length, domain, symbols, etc.) and may not deeply analyze webpage content, images, or embedded scripts.
6. **Scalability Challenges:** Handling a large number of user requests simultaneously may affect performance if the backend is not optimized or deployed on scalable infrastructure.
7. **Dependency on External APIs:** If the system uses third-party services its performance and reliability may depend on those external services.
8. **User Input Dependency:** The system requires users to manually enter or paste URLs, which means it cannot proactively protect users unless integrated with browsers or email systems.

CHAPTER 7: CONCLUSION

The PhishShield AI – Intelligent Phishing Detection and Prevention System is designed to address the growing threat of phishing attacks in today's digital environment. Phishing attacks continue to evolve and pose serious risks to individuals and organizations by attempting to steal sensitive information through deceptive emails, websites, and online communication. The proposed system provides an effective solution by allowing users to analyze suspicious URLs and email content through an intelligent detection platform.

The system integrates artificial intelligence techniques, blacklist verification, and web-based technologies to identify potential phishing threats and generate risk scores that help users understand the severity of malicious content. By providing features such as phishing scanning, report generation, AI chatbot assistance, and administrative monitoring, the system improves both cybersecurity awareness and threat detection capabilities.

The use of UML modelling techniques, including use case, activity, sequence, and class diagrams, helped design a structured and scalable architecture for the system. These diagrams clearly illustrate system workflows, component interactions, and relationships between system modules, supporting efficient system development and maintenance.

Overall, the PhishShield AI system demonstrates how intelligent detection mechanisms and structured system design can significantly enhance protection against phishing attacks. With further improvements and integration of advanced security technologies, the system has the potential to evolve into a comprehensive cybersecurity platform that contributes to a safer digital environment.

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